AMERY ICE SHELF BOREHOLE MICROCAT AND CTD DATA, 2005/2006 SEASON - DATA PROCESSING AND QUALITY

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INTRODUCTION

This report summarises the latest AMISOR oceanographic data, season 2005/2006, collected from oceanographic equipment deployed through the Amery Ice Shelf boreholes, including details of data processing and data quality. Temperature/salinity/pressure time series data were downloaded from two strings of Seabird SBE37IM microcats at AMISOR boreholes AM02 (a year of data) and AM04 (5 days of data, new deployment). CTD casts, using an FSI 3" MicroCTD, were taken through new boreholes at AM03 and AM04. Two additional calibration casts using the FSI CTD were done on Aurora Australis cruise au0603.

Table 1: Station details for AMISOR season 2005/2006 borehole microcats. Note that borehole absolute locations change in time, as the ice shelf is in motion. For the microcats, positive clock errors indicate microcat clocks were fast. Instrument pressures/depths are mean values over the data dates.

Borehole microcats

borehole	position at given date		microcat	instrument pos.		data dates		rec.	clock
	latitude	longitude		depth (m)	pressure (dbar)	start	end	interval (min.)	error (sec.)
AM02	xx°xx.xx'S	xx°xx.xx'E	1623	333.2	336.9	27-Dec-2004	21-Feb-2006	30	+124
AM02	(on xx-xxx-xxxx)		1624	555.1	561.5	27-Dec-2004	21-Feb-2006	30	+300
AM02	"	"	1174	762.0	771.2	27-Dec-2004	21-Feb-2006	30	+125
AM04	69°53.97'S	70°17.42'E	1972	544.1	550.4	11-Jan-2006	16-Jan-2006	30	0
AM04	(on 11-Jan-2006)		1973	684.8	693.0	11-Jan-2006	16-Jan-2006	30	0
AM04	"	"	1974	804.9	814.7	11-Jan-2006	16-Jan-2006	30	0

MICROCAT DATA

The microcat moorings have already been described in previous reports. The mooring at AM02 was deployed in January 2001, while the new mooring at AM04 was deployed in January 2006. An additional new mooring was deployed at borehole AM03, but data were not downloaded. The old mooring at borehole AM01, deployed in January 2002, was visited but not downloaded. All latest microcat data downloaded during the 2005/2006 season (Table 1) were assigned the same decimal time scheme consistent with previous data i.e. decimal days as counted from midnight on December 31st 2000 (so, e.g. midday on January 1st 2001 = 0.5 decimal time; midday on January 1st 2002 = 365.5).

The microcats were downloaded by Dave Rasch using the SeaBird terminal program Seaterm, and instrument clock errors were noted at the time (Table 1). Clocks were reset to GPS time for both sets of microcats, following data download and prior to reinitialisation of logging.

Manufacturer supplied calibrations (July/August/September 2000 for AM02 instruments, February 2003 for AM04 instruments) were applied internally by the microcats, and calibrated data were output. The data were then processed, as follows:

* The files were padded at the start and end, data gaps were checked for and filled, and decimal times were calculated. For microcat 1623, there was a near total break in the data from 5th to 21st June 2005 (reason unkown).

* For AM02 microcats, decimal times were “compressed” linearly throughout the time series to correct for clock error. No significant clock error was noted for the short time series at AM04.

* Data quality control notes:
    - AM02 microcat 1174: temperature spike removed at decimal time 1625.9786
    - AM02 microcat 1174: all conductivity/salinity data removed (sensor went bad in March 2004)

* Microcat data are plotted in Figures 1 and 2. For AM02 microcat 1623, a span of low conductivity/salinity data occurs from decimal time ~1720 to ~1739. It is not known whether these low values are due to fouling, or to some "event" near the base of the ice shelf. These data have not been removed from the files.

* For AM02 and AM04, a mean pressure value was calculated for the year's data for each microcat (Table 1).

* From conductivity and temperature sensor calibration drift estimates in the previous data report for 2003/2004 data (i.e. 0.0003 (PSS78) **per month** for salinity, and 0.0005^{o}C **per year** for temperature), we might expect the following calibration drifts at the time of the latest 2005/2006 data downloads:

AM02 (in water for 5 years) total S drift = 0.018 (PSS78) total T drift = 0.0025^{o}C

For AM04, manufacturer sensor calibrations were from February 2003, and the following total drifts might be expected by the time of deployment in 2006:

AM04 (after 3 years) total S drift = 0.010 (PSS78) total T drift = 0.0015^{o}C

Pressure drift for the AM02 microcats is of the order 0.5 dbar over the latest year of data, decreasing with time for all 3 microcats. This is well within the range of expected pressure sensor drift as quoted by the manufacturer, however from the microcat data alone it is difficult to separate the contributing factors to this drift i.e. calibration drift, and/or physical change in instrument position due to ice shelf change.

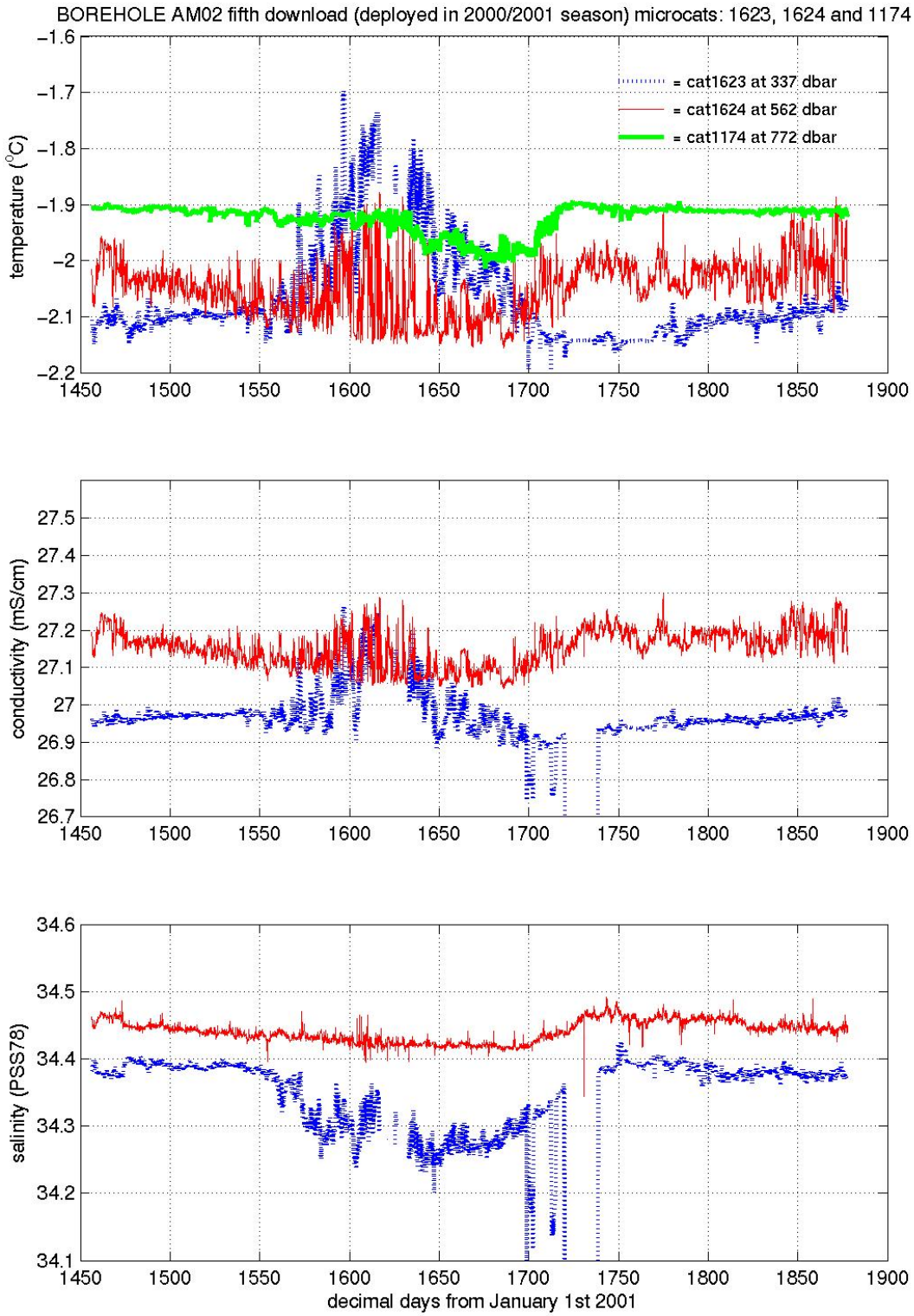


Figure 1: **Borehole AM02 microcat data downloaded in 2005/2006 season.**

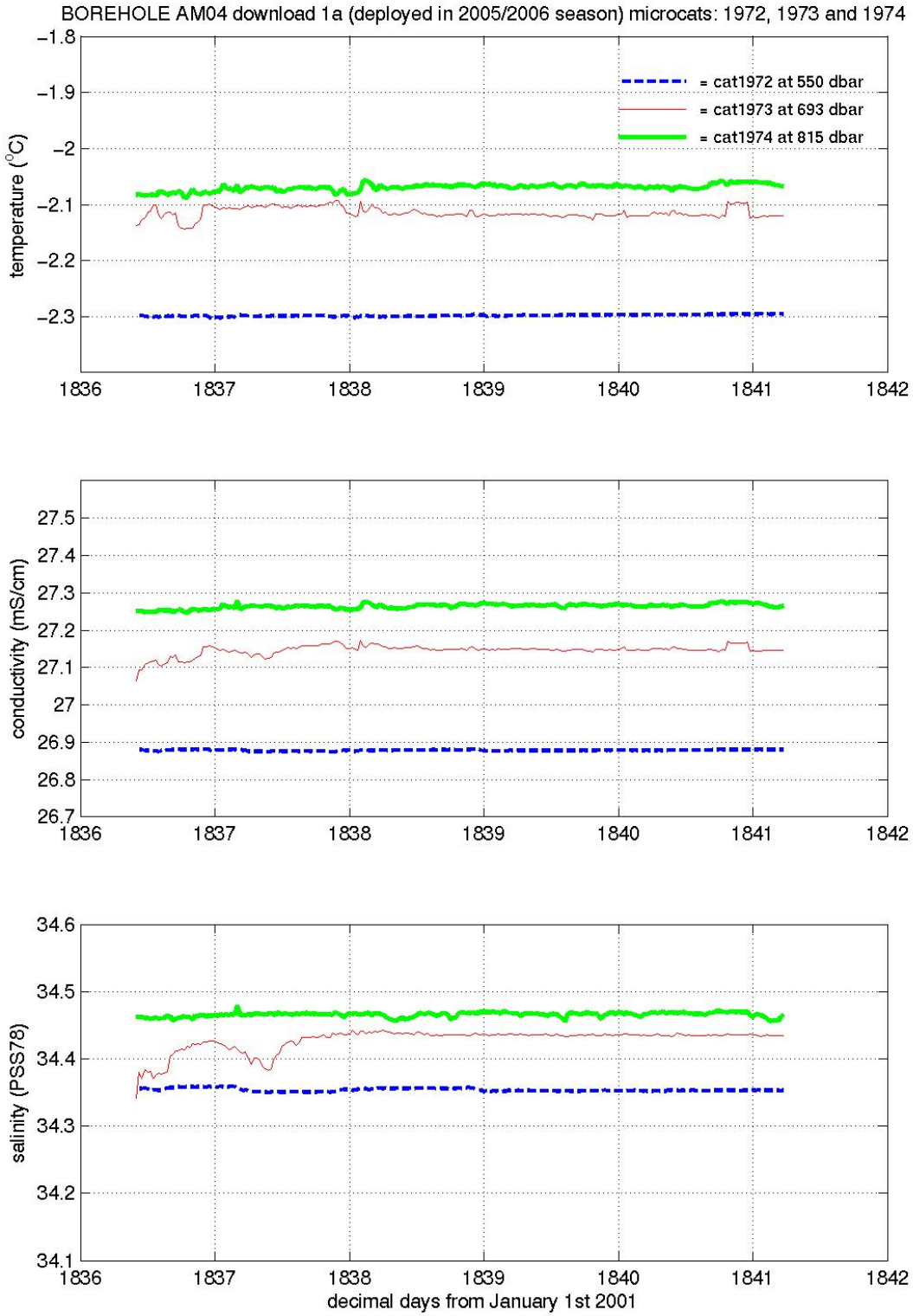


Figure 2: Borehole AM04 microcat data downloaded in 2005/2006 season.

CTD DATA

Pre-season laboratory calibrations of the FSI CTD temperature, pressure and conductivity sensors were done at CSIRO Division of Marine Research (August to September 2005). In the field, "raw" data were output from the CTD in engineering units, with CSIRO calibration coefficients applied for temperature, pressure and conductivity. The bottom contact switch was not fitted for the borehole work. At borehole AM03, the data were output from the instrument at 1.83 Hz and logged direct to PC via the seacable; for borehole AM04 and on cruise au0603, the data were logged internally by the instrument at 0.83 Hz, with data download following each deployment. Further corrections for pressure and conductivity were obtained from 2 casts taken aboard Aurora Australis cruise au0603 (Figure 3). For these 2 casts the FSI MicroCTD, in internally recording battery-powered mode, was attached to the ship's main rosette system, and 2 routine 22 bottle casts were taken (Table 2) with SBE (i.e. Seabird SBE9plus) CTD serial 704. FSI and SBE CTD data were then compared, and FSI conductivity data were calibrated against the salinity Niskin bottle samples obtained. Salinity samples obtained from Niskin bottles deployed through the ice shelf were unusable, so no further in situ conductivity correction for the borehole work was possible.

Communication problems resulted in numerous corrupted data scans, however most of the data were retrievable - manual editing was required.

Table 2: CTD station details for Amery Ice Shelf Boreholes AM03 and AM04, and Aurora Australis cruise au0603 FSI calibration CTD's. Note the following: depth to water surface=distance from top of borehole down to water surface in the borehole; bottom depth=total water depth from water surface to ocean bottom; max.P=maximum pressure of CTD cast; elevation=CTD elevation above bottom at the bottom of the cast.

Borehole CTD

bore -hole	cast no.	UTC start time	UTC start date	latitude	longitude	borehole depth (m)	depth to water surface (m)	bottom depth (m)	max.P (dbar)	elev. (m)
AM03	1	2112	12-Dec-2005	70°33.67'S	70°19.93'E	722	74	1265	1238	43
AM03	4	1631	15-Dec-2005	70°33.67'S	70°19.93'E	722	74	1265	1238	69
AM04	1	1807	09-Jan-2006	69°53.97'S	70°17.42'E	603	64	938	934	16

au0603 CTD

	119	0737	03-Mar-2006	66°04.97'S	79°27.91'E			1497	1552	10
	120	0923	03-Mar-2006	69°05.11'S	79°27.58'E			1476	1531	12

au0603 FSI CTD processing and calibration

The following processing steps were followed for the 2 au0603 casts to obtain calibration corrections for the FSI pressure and conductivity:

* Surface pressure offset was found by averaging the 20 pressure points previous to the CTD entering the water. This offset was then removed from FSI pressure data.

* Upcast burst data were formed by retaining 12 sec. of data previous to each bottle firing, then averaging these 12 sec. bursts. Burst averages were then merged with SBE upcast burst averages, and salinity bottle data.

* Separate pressure monotonic files for downcast and upcast data were formed from the raw data.

* Comparison of FSI and SBE pressure data revealed a small calibration difference, of the order 1.5 dbar over 1600 dbar. Assuming the SBE pressure as a baseline, upcast pressure burst averages for

the FSI CTD were linearly fitted to the SBE pressure burst averages (Figure 4). The following linear correction was then applied to all FSI pressure data:

$$p_{cal} = 1.0008\ p_{raw} - 0.0088 \qquad \text{(eqn 1)}$$

where p_{cal} and p_{raw} are respectively the corrected and uncorrected FSI pressure. Note that this correction was obtained from upcast data only, and no account was taken of FSI pressure sensor hysteresis which may cause slight variation to the downcast calibration.

* FSI conductivity was calibrated using the salinity bottle data (Figure 5), as per the method referenced in previous reports. The 2 stations were grouped together to provide a single calibration fit, however from Figure 5 it is evident that the FSI conductivity cell behaved differently on the 2 casts, equivalent to a salinity difference of the order 0.01 (PSS78). This is a serious problem, and certainly limits the accuracy of the FSI salinity data. This is discussed more later in the report. The linear conductivity correction obtained from the shipboard calibration was

$$c_{cal} = 0.99336\ c_{raw} - 0.031347 \qquad \text{(eqn 2)}$$

where c_{cal} and c_{raw} are respectively the corrected and uncorrected FSI conductivity. This correction was applied to all FSI conductivity data.

* 2 dbar averages were formed for temperature, corrected pressure and corrected conductivity from the pressure monotonic downcast and upcast files. Due to the low data sampling rate, the minimum attendance required to form each 2 dbar bin was only 1 data point. A salinity value for each 2 dbar bin was calculated from the 2 dbar averages. The data are plotted in Figure 3.

Borehole FSI CTD processing and calibration

FSI CTD data from the 2 borehole locations were processed as above for the shipboard data, with the pressure and conductivity corrections (eqns 1 and 2) applied. Note that the following "yoyo's" were removed prior to forming the pressure monotonic files:

* AM03 cast 4, yoyo during upcast between ~720 and 630 dbar
* AM04 cast, yoyo during upcast between bottom and ~535 dbar

Removal of these yoyo's results in a small data discontinuity within the casts.

Complete data files (i.e. yoyo's not removed; no 2 dbar averaging) were also produced for each cast, with the pressure and conductivity corrections applied.

Data quality

Data comparisons between the FSI and SBE CTD's for the 2 shipboard calibration casts on cruise au0603 are shown in Figures 6 and 7. In general, the larger spikes in the difference plots are an expected artefact of pressure sensor and water parcel mismatches in steep vertical gradients.

Temperature

Overall for these downcast data, temperature values between the 2 CTD's agree to within ~0.003^{o}C, usually less (Figures 6a and 7a). Differences are typically greater (~0.002^{o}C) at the lower temperatures, in particular the subzero temperature layer between ~100 and 250 dbar (see Figures 3a and b). Better agreement (close to 0.001^{o}C) is found at higher temperatures.

Downcast and upcast borehole temperature data (Figures 8a, b and c) are consistent, and the temperature data are considered reliable.

Salinity

Salinity differences (Figures 6b and 7b) confirm the significant problems with the FSI conductivity sensor. The magnitude of the difference values for the 2 stations are themselves different (e.g.~ -

0.005 PSS78 below 800 dbar station 119, ~ +0.008 PSS78 below 800 dbar for station 120). This is due to the inconsistent conductivity measurements for the 2 stations, seen from the in situ conductivity calibration (Figure 5). It is possible the physical mounting of the FSI on the ship's rosette frame was altered slightly between the 2 casts, changing the physical interference with the field around the FSI conductivity cell. Alternatively, if the physical mounting was not altered, the FSI conductivity cell is not performing to the required level of repeatability. Either way, the 2 shipboard calibration casts only provide a conductivity correction accurate to ±0.01 (PSS78).

The salinity difference profiles change above ~600 dbar (Figures 6b and 7b). This coincides with increased vertical salinity gradients (Figures 3b and d), and is most likely due to sensor mismatch with the SBE conductivity, and poor flushing of the FSI conductivity cell - the instrument was mounted in a vertical position in between the Niskin bottles, ~0.4 m above the SBE sensors, and not in the optimal position for CTD sensor measurements.

Further evidence for unstable FSI conductivity measurements comes from the borehole work. For AM03 cast 1 and AM04 (Figures 8a and c), there is a suspiciously large difference between downcast and upcast salinity values. AM03 cast 4 salinity data on the other hand shows good agreement between downcast and upcast values. Taking the particular case of the salinity difference around 550 dbar for the AM04 cast (Figure 8c), 5 days' of microcat data at the same depth (i.e. microcat 1972 in Figure 2) show no evidence of this salinity variability. From previous seasons' borehole measurements, good salinity downcast and upcast agreement was found for seasons 2000/2001 and 2001/2002, however a large salinity upcast/downcast difference is present for the single cast from the 2003/2004 season (see previous seasons' data and data reports).

Combining all these potential conductivity errors, in particular the suspect differences between downcast and upcast borehole measurements, the salinity data can only be considered accurate to 0.05 (PSS78). No usable salinity Niskin bottle samples were obtained from the borehole work, thus no further improvement on the accuracy is possible for the current season's data. For future seasons, using the FSI CTD it appears essential that good Niskin bottle samples be obtained for every CTD cast.

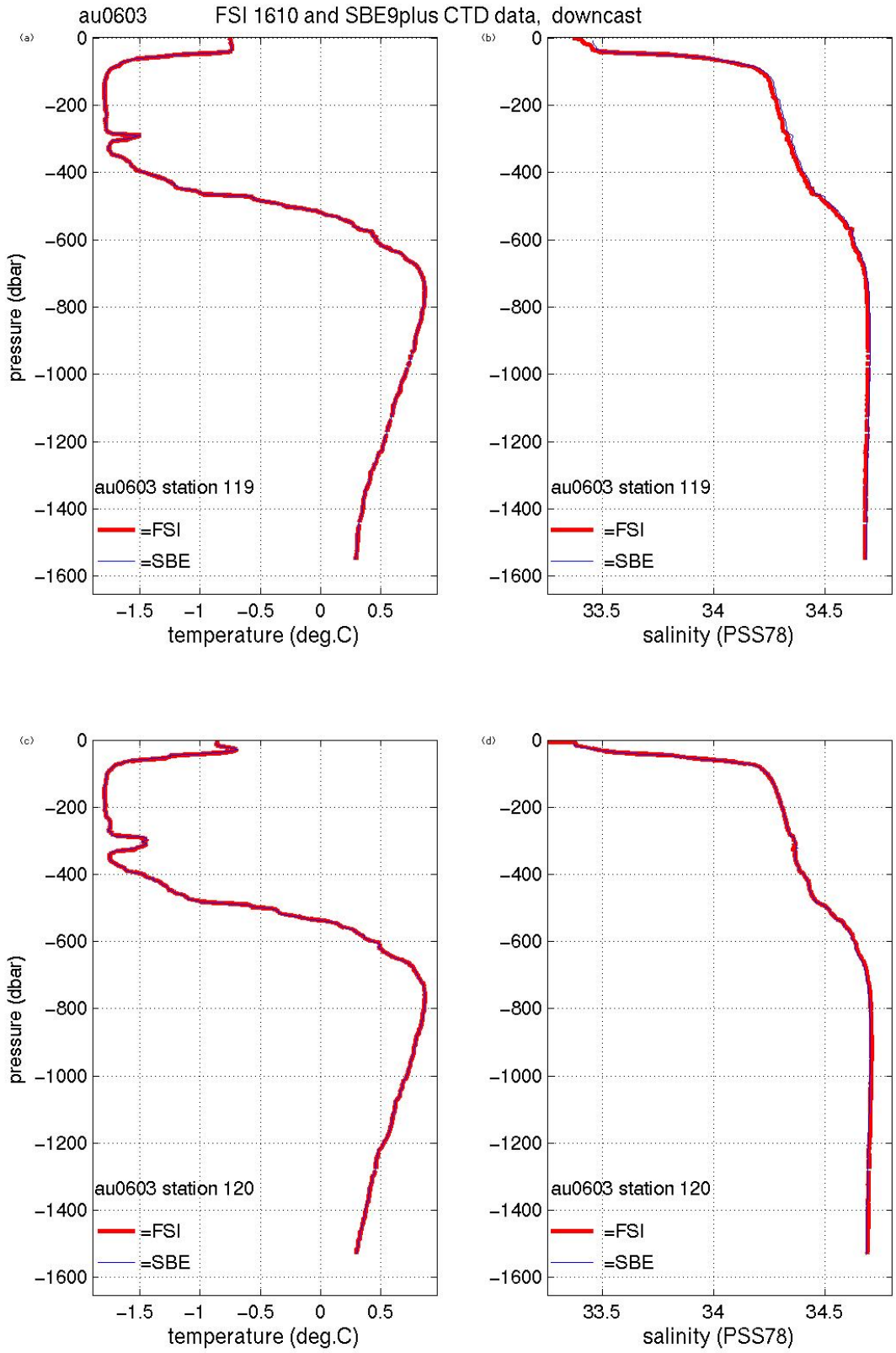


Figure 3: Comparison of FSI and SBE (i.e. Seabird) CTD data, cruise au0603, stations 119 and 120. Downcast data only.

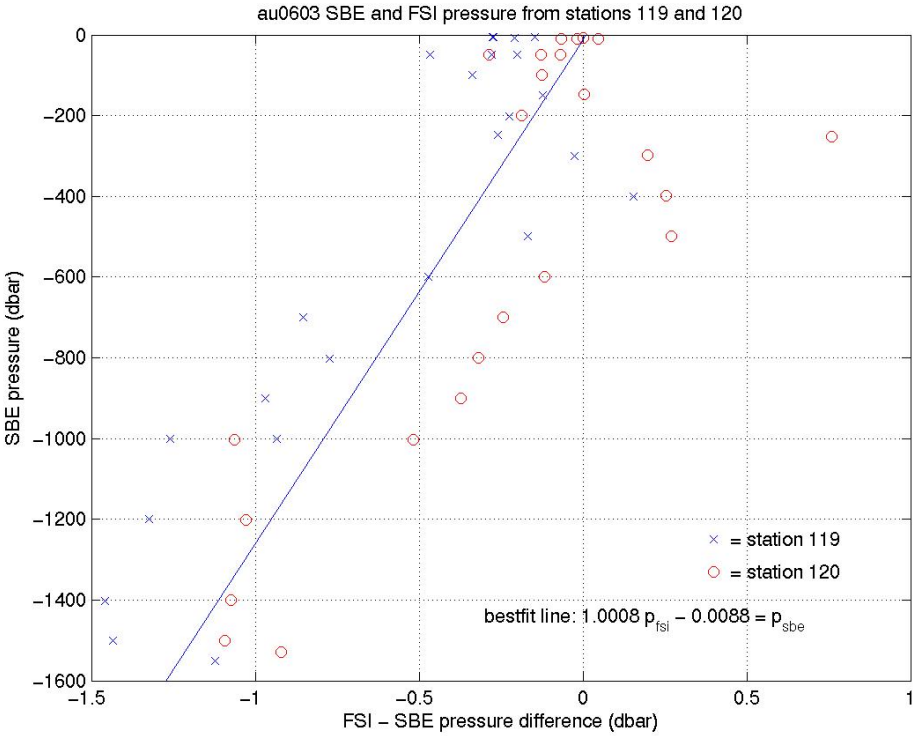


Figure 4: Comparison of FSI and SBE pressure data from Niskin bottle stops on upcast. Bestfit line shows correction to apply to FSI pressure data.

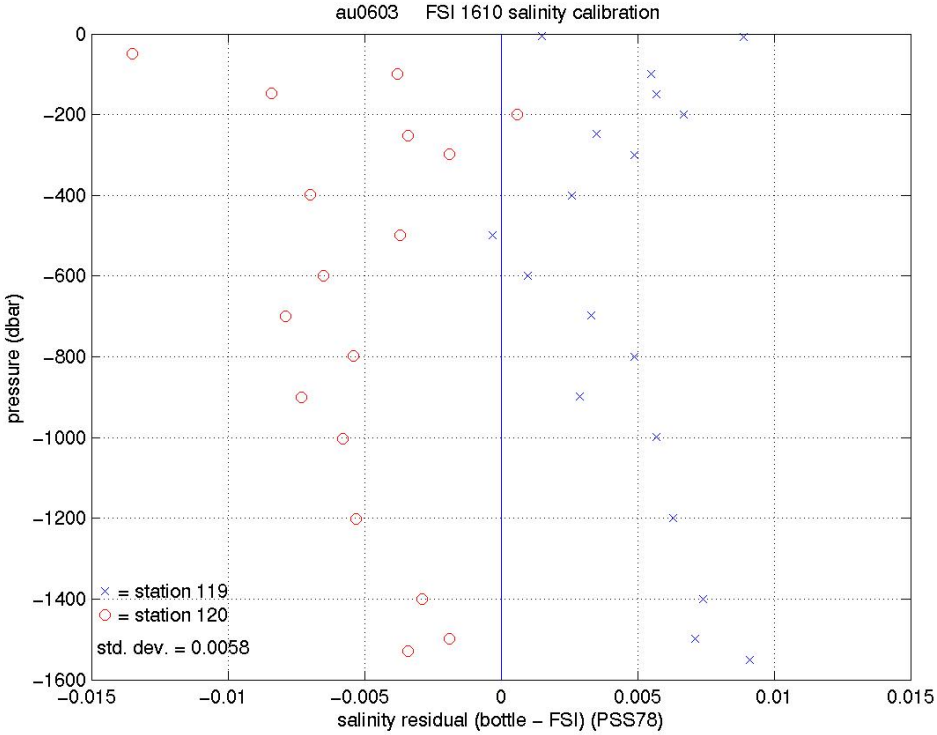


Figure 5: Calibration result for FSI salinity, following calibration against Niskin bottle salinity samples, cruise au0603 stations 119 and 120.

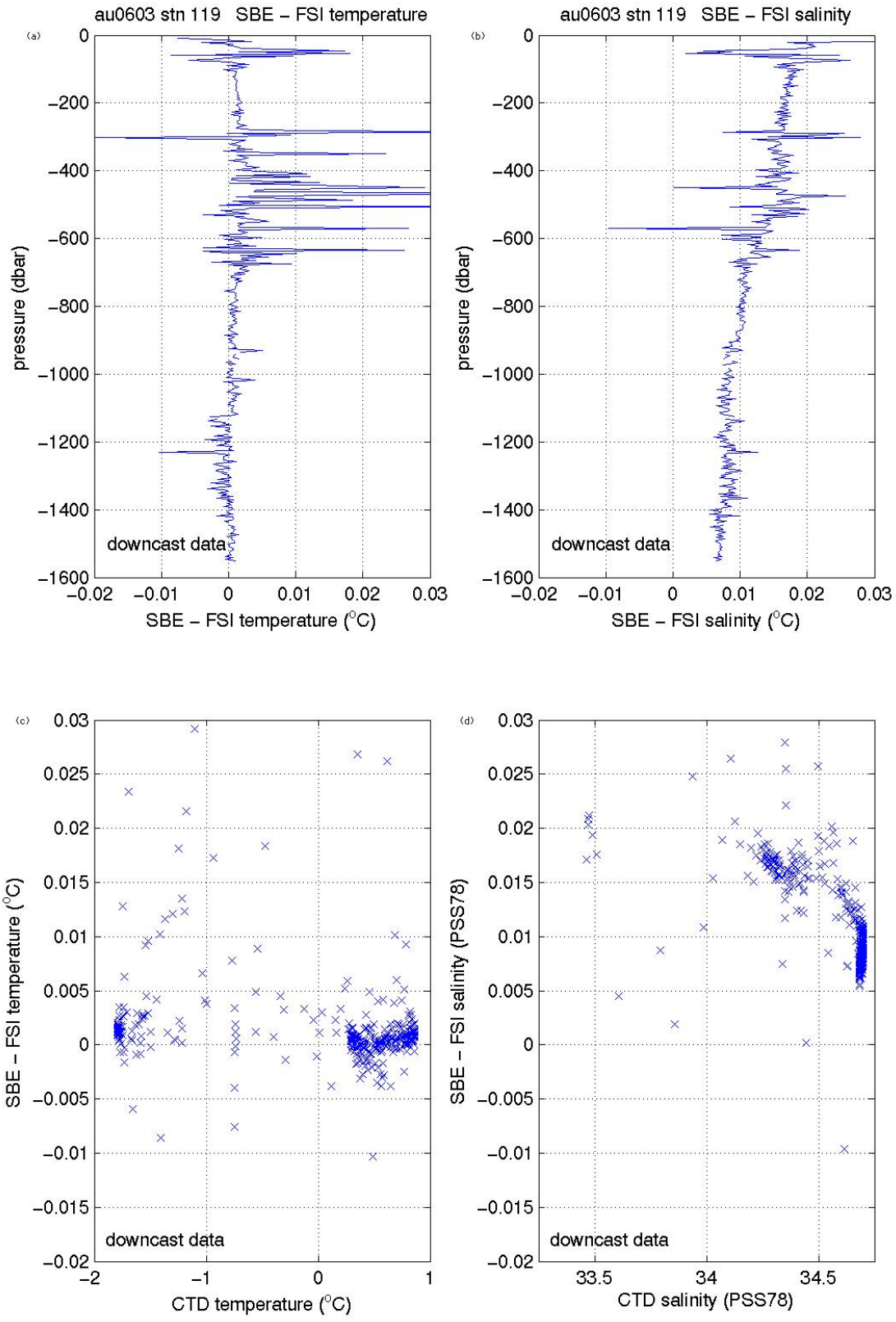


Figure 6: Difference between SBE and FSI CTD temperature and salinity data, cruise au0603 station 119, downcast data.

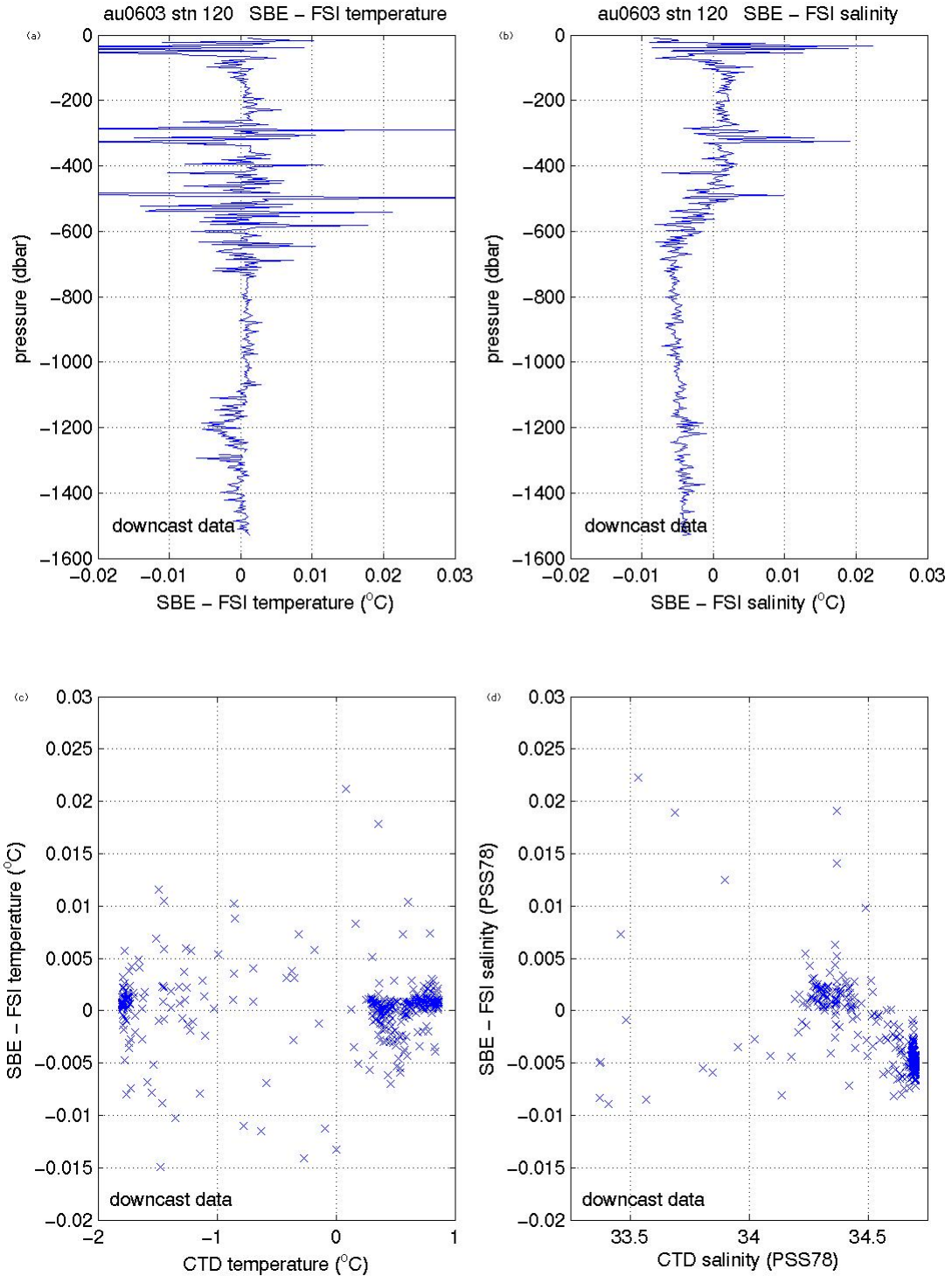
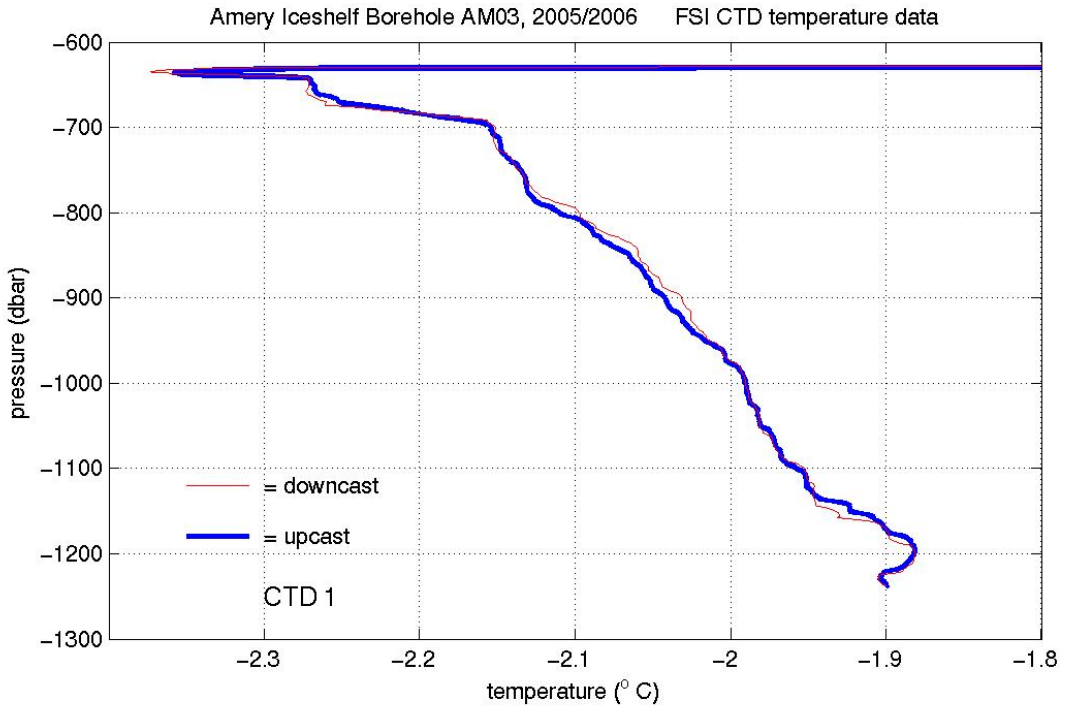


Figure 7: Difference between SBE and FSI CTD temperature and salinity data, cruise au0603 station 120, downcast data.



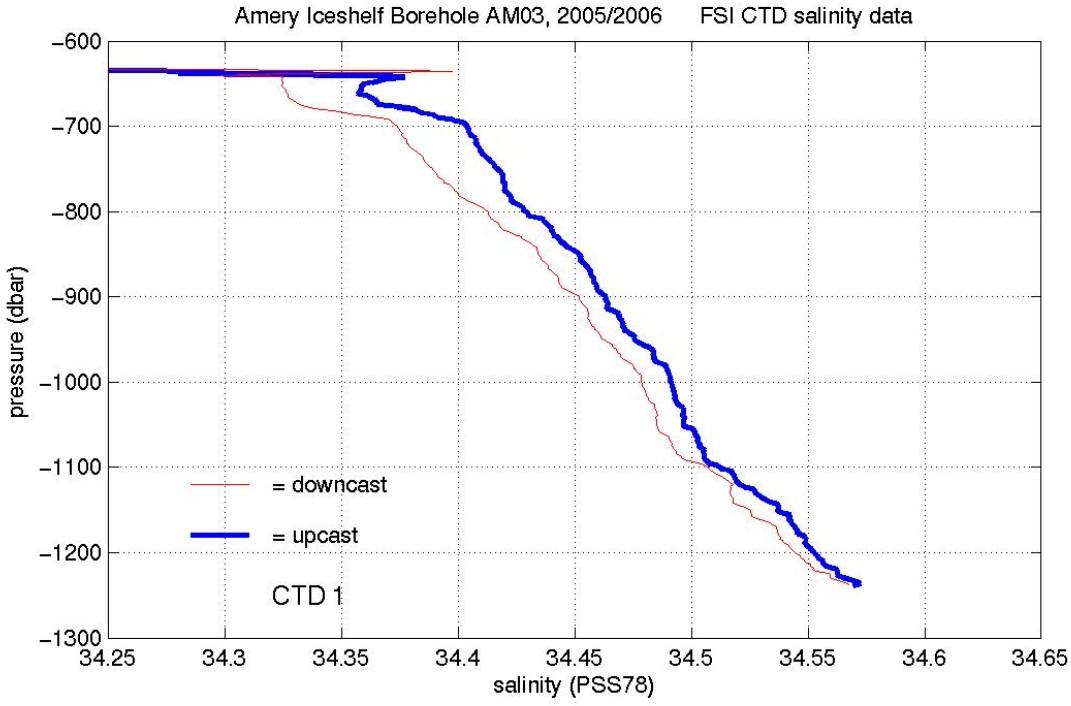


Figure 8a: Amery Ice Shelf borehole AM03, CTD 2 dbar averaged data for cast number 1.

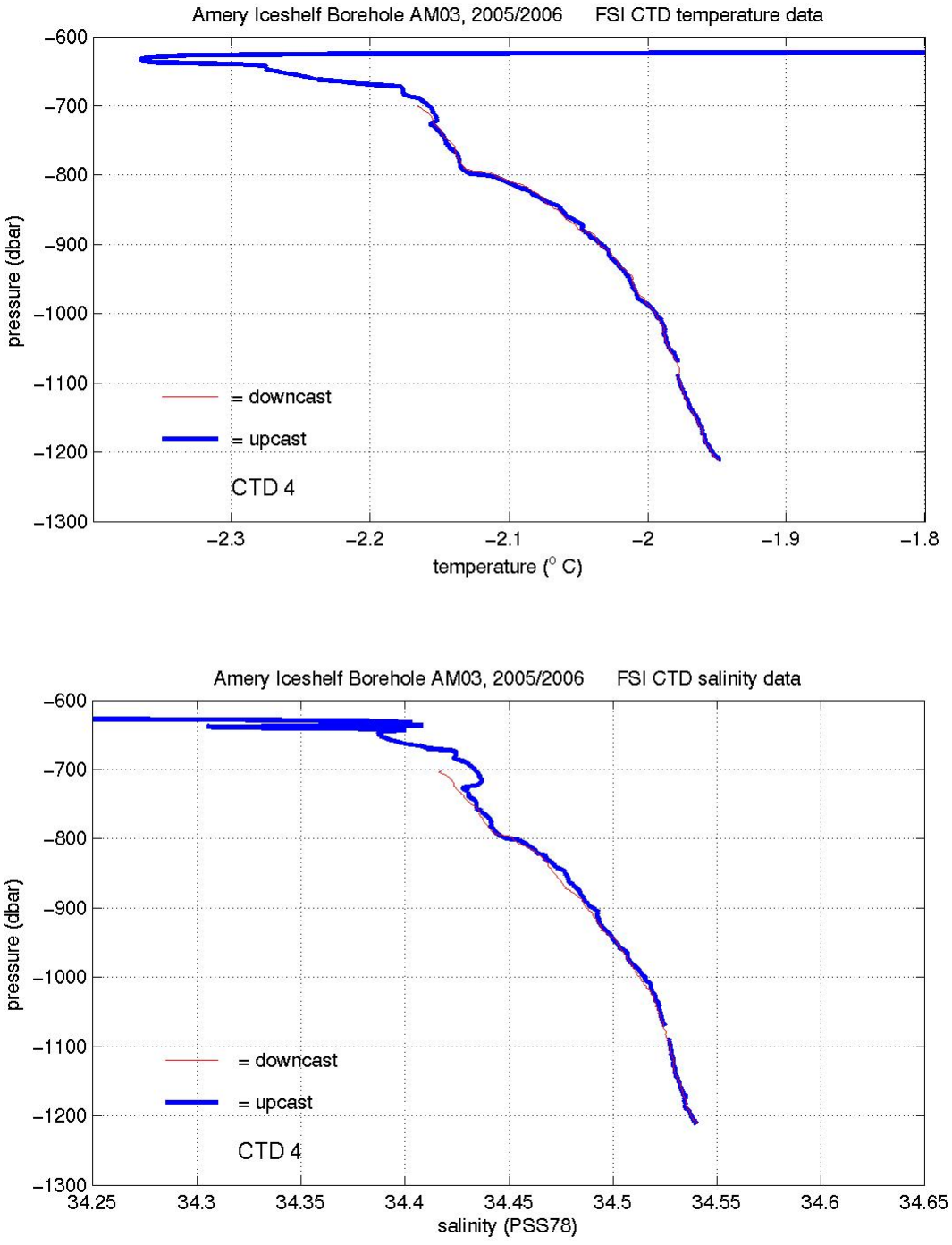
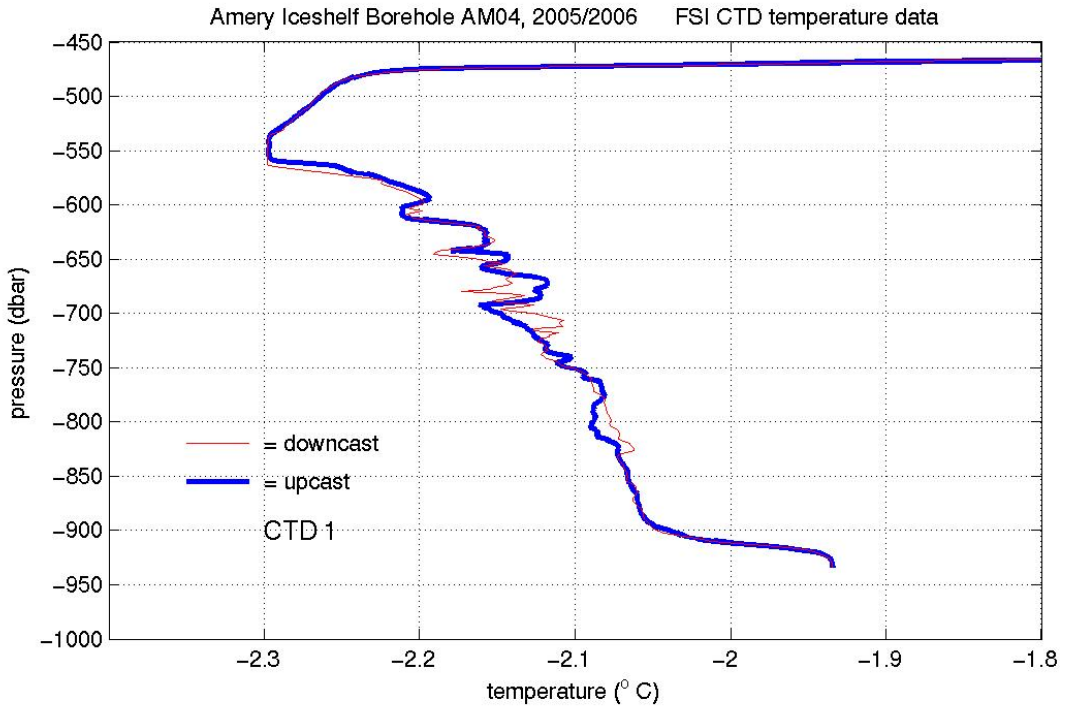


Figure 8b: Amery Ice Shelf borehole AM03, CTD 2 dbar averaged data for cast number 4.



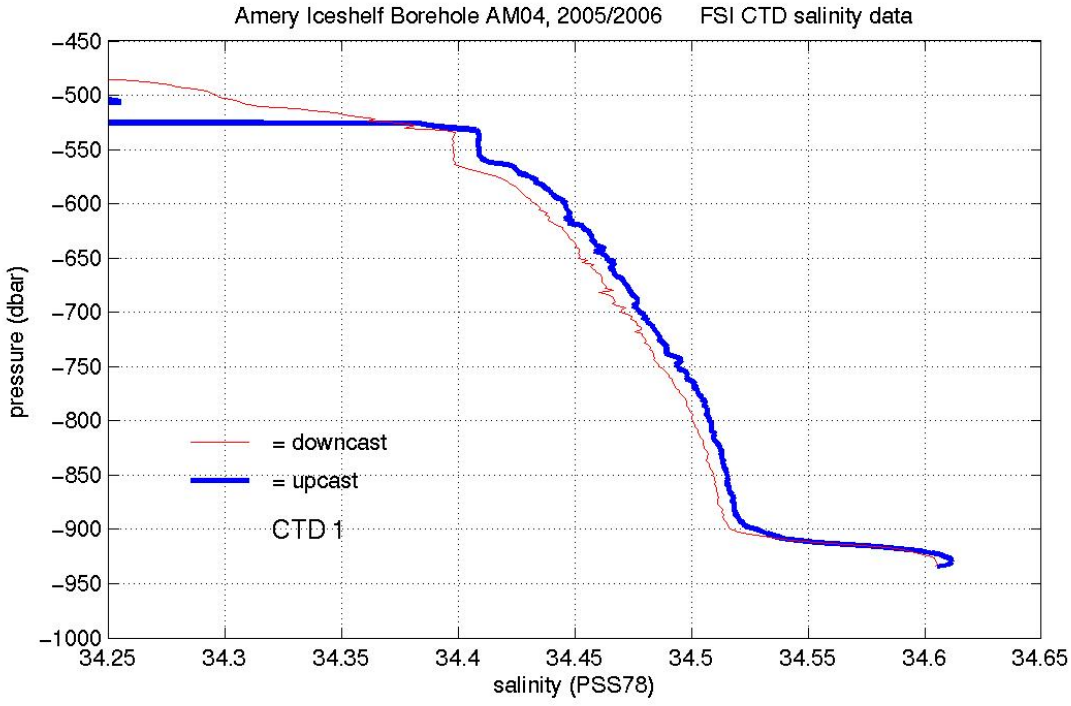


Figure 8c: Amery Ice Shelf borehole AM04, CTD 2 dbar averaged data.

DATA FILES

New microcat data from boreholes AM02 and AM04, and 2 dbar averaged CTD data from boreholes AM03 and AM04, are contained in text and matlab format files, as follows:

1. Microcat data

AM02
Data from the fifth data download (i.e. 27/12/2004-21/02/2006) are contained in the files:
cat1623_5.dat, cat1624_5.dat, cat1174_5.dat (text files), and
cat1623_5.mat, cat1624_5.mat, cat1174_5.mat (matlab files).
READMEcatam02_5 details the microcat file formats.

The updated unified files (i.e. all data from initial deployment time on 06/01/2001, updated with data from the fifth download) are contained in the files:
cat1623_all.dat, cat1624_all.dat, cat1174_all.dat (text files), and
cat1623_all.mat, cat1624_all.mat, cat1174_all.mat (matlab files).
READMEcatam02_all details the microcat file formats.

AM04
Data from the first download (download 1a) (i.e. 11/01/2006-16/01/2006) are contained in the files:
cat1972_1a.dat, cat1973_1a.dat, cat1974_1a.dat (text files), and
cat1972_1a.mat, cat1973_1a.mat, cat1974_1a.mat (matlab files).
READMEcatam04_1a details the microcat file formats.

2. CTD data

AM03
Data from the 2 usable CTD casts at borehole AM03 are contained in the files:

fsiam0300x.dwc_av	cast x (x=1 or 4) downcast 2 dbar averaged data, text files
fsiam0300x.upc_av	cast x (x=1 or 4) upcast 2 dbar averaged data, text files
fsiam03.mat	downcast and upcast data for both casts, matlab file
fsiam0300x.ful	cast x (x=1 or 4) complete data at the full data rate, text files
READMEam03_ctd	a README file detailing file formats

AM04
Data from the single CTD cast at borehole AM04 are contained in the files:

fsiam04001.dwc_av	downcast 2 dbar averaged data, text file
fsiam04001.upc_av	upcast 2 dbar averaged data, text file
fsiam04.mat	downcast and upcast data, matlab file
fsiam04001.ful	complete data at the full data rate, text file
READMEam04_ctd	a README file detailing file formats

3. Station and summary info files

The files station.dat and borehole_data_summary.dat are updated to include the latest season's data.